ClaimGuard AI — System Architecture & Technical Design

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1. Executive Overview

ClaimGuard AI is an enterprise-grade agentic insurance verification and fraud detection system designed to automate claim intake, prevent duplicate billing fraud, and provide explainable decision dossiers. It integrates IBM Db2 as the ACID transactional system-of-record with an autonomous 5-agent LangGraph workflow and Google Gemini 3.6 Flash multimodal reasoning.

2. IBM Db2 Relational Architecture

IBM Db2 functions as the core persistence engine. The schema comprises 7 tables with strict relational integrity:

Table Name	Primary Key	Key Attributes & Purpose
CUSTOMER_DETAILS	CUSTOMER_ID	First/last name, DOB, National ID, Phone, Email, Address
POLICY_DETAILS	POLICY_ID	Policy type, status, start/end dates, max/remaining coverage, deductibles, co-pay %
CLAIM_RECORDS	CLAIM_ID	Invoice number, hospital name, diagnosis, amount, status, risk score, decision notes
DOCUMENT_RECORDS	DOCUMENT_ID	File path, MIME type, SHA-256 hash, extracted JSON payload (CLOB)
FRAUD_ALERTS	ALERT_ID	Alert type, severity (CRITICAL/HIGH/MED), description, resolution status
VERIFICATION_LOGS	LOG_ID	Agent name, action executed, status, execution time (ms), details (CLOB)
USERS	USER_ID	RBAC credentials, username, PBKDF2 hashed password, roles (REVIEWER/ADMIN)

3. LangGraph Multi-Agent Orchestration

The verification workflow is modeled as a state machine coordinated by LangGraph:

- **Document Extraction Agent:** Ingests clinical invoices and bills. Uses LangChain ChatGoogleGenerativeAI (gemini-3.6-flash) with structured output parsing and deterministic regex fallbacks.
- **Database Verification Agent:** Queries IBM Db2 using parameterized SQL (ibm_db) to verify policy validity, remaining benefit limits, deductibles, and prior claim histories.
- **Fraud Detection Agent:** Executes dual-layer security checks: (1) Cryptographic SHA-256 hash collision matching against DOCUMENT_RECORDS, (2) Duplicate invoice collision detection against CLAIM_RECORDS, and (3) Inconsistent hospitalization dates.
- **Decision Support Agent:** Evaluates composite findings into an objective 0–100 risk score and produces APPROVE, INVESTIGATE, or REJECT recommendations with transparent rationales.

• **Supervisor Agent:** Routes state, coordinates step execution, logs all actions to Db2 VERIFICATION_LOGS, and enforces final human examiner sign-off.

4. Technology Stack & Enterprise Standards

• **Database:** IBM Db2 Community Edition (Docker / Cloud) via native C-bindings ibm_db (v3.3.0).
• **Backend:** FastAPI, Uvicorn, Python 3.14, Pydantic v2 schemas, JWT (HS256) authentication, PBKDF2 hashing.
• **AI Layer:** LangChain, LangGraph StateGraph, Google Gemini 3.6 Flash multimodal model.
• **Frontend:** React 18, Vite, Tailwind CSS, Recharts, Lucide React styled in IBM Carbon design language.
• **Quality Assurance:** 75/75 Pytest passed (100%), 10/10 Frontend tests passed (100%), 0 secrets committed.